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Original Research

Do social frontiers matter for depression?

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ABSTRACT

Social frontiers – abrupt borders between communities – may heighten territorial and defencive behaviour, reduce opportunities for positive contact between groups, and exacerbate the sense of outgroup threat, resulting in a negative impact on mental ill health such as depression for residents living in neighbourhoods bounded by social frontiers. Previous research on the links between residential segregation and depression has largely ignored the effect of social frontiers. To study this association we link Place Based Longitudinal Data Resource data on the numbers of depression diagnoses and antidepressant drugs prescribed by GPs with estimates of ethnic and religious social frontiers produced from the 2011 and 2021 Census for all Lower Super Output Areas in England. The estimated associations are produced from spatial binomial/Poisson regression models that allow for spatial autocorrelation via adding a spatially lagged response term to the model. We find strong and consistent evidence of an association between the prevalence of depression at the neighbourhood level (Lower Super Output Areas) in England and the intensity of social frontiers for particular ethnic (Chinese, Indian, Pakistani, White British) and religious (Hindu, Jewish, Muslim) groups. For example, in 2021 depression rates are estimated to be between 1.5% and 2.2% higher for every 5% point increase in the intensity of social frontiers between Chinese and non-Chinese residents. These results demonstrate the importance of understanding the role of community boundaries when considering the links between segregation and mental wellbeing.

1. Introduction

Social frontiers – abrupt transitions between neighbourhoods in social, ethnic or religious mix – have emerged as a previously overlooked but important aspect of residential segregation (Dean et al., 2019; Piekut et al., 2019; Pryce, 2021; Křížková et al., 2021). The lack of residential mixing along the boundaries between residential groups has previously been shown to have a number of adverse consequences, affecting crime rates (Legewie and Schaeffer, 2016; Dean et al., 2019), residential mobility (Olner et al., 2024), and housing discrimination (Binner et al., 2024). Despite these findings, the potential mental health consequences of living near social frontiers remain largely overlooked in empirical research. The only study we are aware of focuses on Peace Walls in Northern Ireland (Maguire et al., 2016). It finds large and statistically significant negative associations with the take-up of anti-anxiety and antidepressant medications. However, it is unclear whether these findings can be generalised beyond the unique context of Northern Ireland to 'invisible' frontiers — unfortified neighbourhood boundaries prevalent in UK, European and North American cities (Dean et al., 2019; Křížková et al., 2021; Legewie and Schaeffer, 2016; Legewie, 2018).

Beyond the specific concept of social frontiers that motivate this paper, the most developed literature linking residential segregation and mental health is the ethnic density literature. The general finding from such studies is that for individuals from ethnic minority backgrounds, living in areas with a high proportion of one's own group carries a protective effect on mental health (e.g., Bosqui et al., 2014; Das-Munshi et al., 2010). However, this literature has overlooked the potentially negative offsetting effect of living near a social frontier as outlined above. Thus these studies may conflate ethnic minority individuals living in high-density areas that are not in close proximity to social frontiers with those also living in high-density areas that are exposed to social frontiers and the various effects they entail. The implication is that by failing to account for social frontier effects, ethnic density studies may significantly underestimate the protective effect of living near one's own group in the absence of frontiers.

Moving beyond the specific ethnic density literature, the results from other studies linking residential segregation and mental health

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have been somewhat ambiguous (Mair et al., 2008). This may be because of the over-reliance on quantifying the level of segregation among different ethnic groups using the index of dissimilarity D (Duncan and Duncan, 1955), rather than the potentially important effects of social frontiers. Kramer (2017) for example, lamented how *'empirical research on neighbourhood boundary making is practically non-existent'*. This omission is important because while D is a very important measure for quantifying overall segregation across a region, it is not designed to identify the extent of social frontiers between each pair of neighbouring areas (see Section 1 of the supplementary material for an illustration of this point). By examining what happens where neighbourhoods border, we aim to shed new light on the association between segregation and mental health.

For the present study we hypothesise that social frontiers are likely to have a negative association with mental ill health such as depression. This is because social frontiers represent a lack of residential mixing at the interface of contrasting residential communities, reducing the opportunity for positive contact between groups (Iyer and Pryce, 2024) which is recognised as a key mechanism in reducing bias and improving intergroup attitudes (Allport, 1954; Pettigrew, 2008; Pettigrew and Tropp, 2008; Lemmer and Wagner, 2015). There are also territoriality effects. Because social frontiers provide a clear demarcation of ingroup and outgroup residential territories they are more likely to evoke territorial behaviour than blurred or ambiguous boundaries (Sack, 1983; Iyer and Pryce, 2024), which is likely to sour the limited opportunities for intergroup contact. Moreover, social frontiers suggest *'an absence or shortage of "bridgebuilders"– those willing to live at the interface between communities'* and who *'cushion inter-group tensions'* (Dean et al., 2019). The absence of the buffering effect provided by bridgebuilders may exacerbate divisions and further limit opportunities for intergroup contact.

Such divisions will also likely generate selection effects. The emergence of social frontiers suggests an aversion to living near the outgroup. As a result, there will be lower housing demand and lower house prices near the frontier. This means that low-income households are likely to be selected into these areas, either by the market (*'coming to the nuisance'*, Depro et al., 2015), or as a result of social housing allocations (Manley and Van Ham, 2011). The concentration of low income households near social frontiers may heighten the sense of scarcity and competition for resources, as well as the anxiety stemming from outgroup threat (Esses et al., 2001; Iyer and Pryce, 2024). Finally, social frontiers may attract protests and political messaging as activists seek to exploit their ready symbolism. As a result, social frontiers may become preferred locations for demonstrations, marches and acts of terrorism or intimidation (Iyer and Pryce, 2024). This will further exacerbate the sense of insecurity, intrusion and stress for those living in frontier communities.

In summary then, there are good theoretical reasons to believe that social frontiers allow the negative mechanisms that sour intergroup relations to flourish, while preventing the positive mechanisms associated with intergroup contact from emerging. Therefore in this paper we present the first empirical evidence on the association between social frontiers and mental ill health outcomes at the small-area level. Specifically, we focus on social frontiers in relation to both ethnicity and religion, and estimate their impact on the rates of depression in England in 2011 and 2021 at the lower super output area level. The data and study design are presented in Section 2, while the statistical models are described in Section 3. The results of the study are presented in Section 4, while the paper concludes in Section 5 with a discussion of the key findings and future work.

2. Study design

The study is based in mainland England excluding the isles of Wight and Scilly, and we utilise data from the last two census years of 2011 and 2021 to assess the robustness of our results. The study has a small-area spatial ecological design, where England is partitioned into n = 32,754 Lower Super Output Areas (LSOA) that have an average population of 1600 individuals. To address the boundary changes in LSOAs between the two census years, a common set of LSOAs based on the 2011 census are used for both time periods. The data used in this study are described below, and a table summarising the mean, minimum and maximum values for each variable across the n LSOAs is presented in Section 2 of the supplementary material.

2.1. Depression outcomes

Data on LSOA-level prevalence of depression is obtained from the Place Based Longitudinal Data Resource, and are freely available to download from https://pldr.org/. The individuals that make up the data are those who are registered with a GP practice, which is the vast majority of the English population. The prevalence of depression is measured in two ways, which allows the robustness of our results to be assessed. The first is the percentage of the population in each LSOA who have a doctor-confirmed diagnosis of depression (obtained since April 2006), while the second is the number of antidepressants prescribed to people living in each LSOA in a year. Note, that antidepressant usage is a useful complement to the depression rate outcome because they are prescribed not only for depression but also for other related conditions that may be triggered or exacerbated by the stresses of living near social frontiers, such as anxiety disorders, OCD, eating disorders, migraines and chronic headaches. The list of 195 antidepressants that make up the second outcome variable are available at the above website.

2.2. Social frontiers

Social frontiers represent locations where there are large differences between the populations living in geographically neighbouring LSOAs, which contradicts Tobler's first law of geography which states that *'Everything is related to everything else, but near things are more related than distant things'* (Tobler, 1970). We investigate the impact of social frontiers in ethnicity and religion in this study, using data on the percentages of the population in each of the following groups.

Ethnicity - (i) African; (ii) Caribbean; (iii) Chinese; (iv) Indian; (v) Pakistani; (vi) Bangladeshi; and (vii) White British.

Religion - (i) Christian; (ii) Hindu; (iii) Jewish; (iv) Muslim; and (v) none.

A description of how social frontiers are constructed for each areal unit and ethnicity/religion variable is presented in Section 3, while a table summarising the distribution of the percentages in each group across the LSOAs is given in Table 1 in the supplementary material.

2.3. Confounders

To reliably estimate the association between the prevalence of depression and social frontiers, we need to account for confounding in the modelling process. The first set of confounders we adjust for relate to population demography including biological sex, age and ethnicity, because existing research shows that mental ill health varies across these demographic groupings (see for example Nazroo, 2003; McManus et al., 2016). Specifically, we compute the percentage of people in each LSOA who are: (a) female; (b) in the following age groups - 16–24, 25–34, 35–44, 45–54, 55–64, 65–74 and 75+; and (c) in the ethnic groups listed in the previous section. Additionally, we include the percentages of the population in each LSOA who are in the religious groups listed above, so that when estimating the effects of social frontiers that represent differences in these percentages between neighbouring LSOAs we have fairly accounted for the sizes of their percentages within each LSOA.

The second set of confounders are measures of socio-economic deprivation, which are included because the link between poverty and mental ill health is well known (e.g., Propper et al., 2005). In England, LSOA-level socio-economic deprivation is measured by the English Index of Multiple Deprivation (IMD), which is a composite index comprising scores in the domains of crime, education, employment, environment, health, housing and income. The health domain is removed because the outcome variable is health related, resulting in 6 variables for inclusion in the models. These IMD scores relate to the closest years to the censuses with matching LSOA definitions, which are 2015 and 2019 respectively. Finally, median annual property price is considered as a possible confounder, because the above measures of socio-economic deprivation measure the level of poverty rather than affluence, i.e., they can struggle to distinguish between moderately affluent and very affluent communities. A small number of LSOAs (around 1%) had missing property price values, which were imputed by spatio-temporal averaging of neighbouring data values.

The final set of confounders relate to household composition, specifically: (i) the percentage of single person households; and (ii) the percentage of households with dependent children. These variables are included because loneliness or having the stresses of caring responsibilities may worsen an individual's mental health. Additionally, we also consider the population density in each LSOA defined as the number of people per km^2, because more densely populated LSOAs have less individual space and would likely be more at risk from antisocial behaviour such as noise pollution from neighbours (MacCutcheon, 2021).

2.4. Research questions

The main aim of this study is to investigate whether and to what extent living near a social frontier has a deleterious impact on the rate of depression in the population, focusing on the following specific research questions.

1. Is there any evidence that living next to an ethnic or religious social frontier has a deleterious impact on depression rates?
2. Social frontiers with respect to which ethnic and religious groupings are most strongly associated with increased rates of depression?
3. How robust are the results across the two census years and the two outcomes used to measure the prevalence of depression?

3. Methodology

This section outlines the methodology used in this paper, including notation and data definitions in Section 3.1, the construction of the social frontiers in Section 3.2 and the estimation of their effects on depression in Section 3.3.

3.1. Notation and data definition

Partition mainland England into n non-overlapping LSOAs represented by $\mathcal{S} = \{S_1, \ldots, S_n\}$, and let $\mathbf{N} = [N(S_1), \ldots, N(S_n)]^\top$ denote the $n \times 1$ vector containing the total population size in each of these areal units. Then let $\mathbf{Y}^D = [Y^D(S_1), \ldots, Y^D(S_n)]^\top$ and $\mathbf{Y}^A = [Y^A(S_1), \ldots, Y^A(S_n)]^\top$ respectively denote $n \times 1$ vectors containing the total numbers of doctor-confirmed depression diagnoses (D) and antidepressants prescribed (A) for each LSOA in a given year. The $n \times 1$ vector of social frontier estimates is denoted by $\mathbf{x} = [x(S_1), \ldots, x(S_n)]^\top$, and their construction is described in Section 3.2 below. Additionally, let $\mathbf{z}(S_i) = [z_1(S_i), \ldots, z_p(S_i)]_{1\times p}$ denote a vector of p confounders, details of which are outlined in Section 2.3 above. Finally, note, as the analyses are conducted separately for each census, then no time index is included here for notational clarity.

3.2. Social frontier construction

A social frontier represents a large difference between two neighbouring communities in some respect, such as in their ethnic or religious makeup or in their socio-economic status. A number of studies have measured the magnitude of social frontiers in areal unit data, with Legewie and Schaeffer (2016) computing the absolute difference in some population characteristic (e.g., ethnicity) between neighbouring areas, which they term *edge intensity*. In contrast, Dean et al. (2019) utilise a binary definition of social frontiers, with a frontier being identified if the edge intensity between two areal units exceeds some threshold. We follow the former absolute difference approach here, so that we can distinguish between social frontiers of different sizes (edge intensities). However, the aim of our study is not to measure the size of each social frontier, but rather to quantify what impact living in an area with larger social frontiers has on poor mental health. For this reason we need to extend the edge intensity concept of Legewie and Schaeffer (2016) by computing a representative frontier size for each areal unit.

For a single ethnic or religious grouping let $\mathbf{x}^* = [x^*(S_1), \ldots, x^*(S_n)]^\top$ denote the $n \times 1$ vector containing the percentages of the population in each LSOA who belong to that grouping. Then we measure the size of the social frontier around LSOA S_i with respect to this variable by considering the set of absolute differences between $x^*(S_i)$ and its spatial neighbours $\{x^*(S_j); j \sim i\}$, where $j \sim i$ denotes that $\{S_i, S_j\}$ share a common border. We measure the overall size of the social frontier around LSOA S_i by either the mean or the maximum of its frontiers across all geographical neighbours, that is by computing:

$$\textbf{Mean frontier size} \rightarrow \quad x(S_i) = \frac{1}{m_i} \sum_{j\sim i} |x^*(S_i) - x^*(S_j)|. \tag{1}$$

$$\textbf{Maximum frontier size} \rightarrow \quad x(S_i) = \max \left\{ |x^*(S_i) - x^*(S_j)|; j \sim i \right\}. \tag{2}$$

Here, m_i denotes the number of LSOAs that share a border with S_i. We consider both the mean and maximum frontier size metrics here as a robustness analysis to assess the sensitivity of our conclusions to the construction of the social frontiers. Note, in this study we include the social frontier metric relating to each ethnic and religious grouping in a separate model so as not to potentially confound the effects of one frontier type with another.

We highlight that the metrics (1) and (2) measure the size of the social frontiers in the ethnic or religious make up surrounding each areal unit, and we aim to identify what impact these social frontiers have on mental ill health. A related but subtly different area of research that is important to draw links to uses statistical models to identify the locations of discontinuities (frontiers) in the health outcome, and a range of approaches have been proposed using graph theoretic ideas (Lee et al., 2021), a second level covariate regression model (Lee and Mitchell, 2012), a spatial clustering prior (Knorr-Held and Raßer, 2000) or a mixture model (Aregay et al., 2016). Collectively, this literature highlights that the spatial correlation structure typically observed in areal unit data is not globally smooth, but instead has sub-regions of smoothness separated by discontinuities (frontiers), and a recent discussion of this issue is provided by Griffith (2023).

3.3. Estimation of social frontier effects

The depression and antidepressant outcomes are modelled separately to each other and separately for each census year, resulting in four sets of model results. Both outcomes $\{Y^D(S_i), Y^A(S_i)\}$ take the form of counts, of the numbers of depression diagnoses or antidepressants prescribed for the population living in LSOA S_i. In the case of depression diagnoses each individual can only be diagnosed once or not at all, while for the antidepressants outcome each individual may have been prescribed multiple courses of antidepressants within the census year. This means that a binomial logistic regression model is appropriate for depression diagnoses, while a Poisson log-linear

regression model is suitable for antidepressant prescriptions. Thus, we adopt overdispersed versions of the following data likelihood models for the two mental ill health outcomes, where the overdispersion allows the variation in the data to be greater than that allowed under the rigid mean–variance relationship specified by the models below.

Depression: $Y^D(\mathcal{S}_i) \sim \text{Binomial}[N(\mathcal{S}_i), \theta^D(\mathcal{S}_i)]$ and $\ln[\theta^D(\mathcal{S}_i)/\{1 - \theta^D(\mathcal{S}_i)\}] = \eta^D(\mathcal{S}_i)$.

Antidepressants: $Y^A(\mathcal{S}_i) \sim \text{Poisson}[N(\mathcal{S}_i)\theta^A(\mathcal{S}_i)]$ and $\ln[\theta^A(\mathcal{S}_i)] = \eta^A(\mathcal{S}_i)$.

Here, $\theta^D(\mathcal{S}_i)$ denotes the estimated proportion of individuals in LSOA $\mathcal{S}_i$ who were diagnosed with depression, while $\theta^A(\mathcal{S}_i)$ denotes the rate, relative to $N(\mathcal{S}_i)$, of antidepressant usage. The linear predictors $\{\eta^D(\mathcal{S}_i), \eta^A(\mathcal{S}_i)\}$ corresponding to these proportions/rates are modelled using the same model structure, which after removing the (D, A) superscripts for notational clarity is given by

$$\eta(\mathcal{S}_i) = \beta_0 + x(\mathcal{S}_i)\beta + \mathbf{z}(\mathcal{S}_i)^\top \boldsymbol{\delta} + \alpha \sum_{j=1}^{n} w_{ij} Y^*(\mathcal{S}_j). \quad (3)$$

The linear predictor is modelled by four components, comprising an intercept term β_0, the effect of the social frontier controlled by β which is the primary inferential target, the effects of the confounders controlled by $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_p)$, and a final term that is included to capture any remaining residual spatial autocorrelation in the data after adjustment for the covariates. Existing spatial association studies such as Haining et al. (2010) and Lee et al. (2022) have modelled this latter term with a set of spatially autocorrelated random effects represented by a conditional autoregressive prior, but Hodges and Reich (2010) and Reich et al. (2021) have shown that this can have deleterious effects on the parameter estimates of spatially autocorrelated fixed effects (e.g., β). Therefore, here we adopt the approach utilised by Lee and Neocleous (2010), which accounts for this spatial autocorrelation by adding a spatially lagged response term on a transformed scale to the linear predictor. We note that the structure of this term has parallels with simultaneous autoregressive (SAR) models, and a review of such models is given by Haining and Li (2020). The specification of the transformed response $Y^*(\mathcal{S}_j)$ depends on the data likelihood model, and is transformed in the binomial and Poisson cases to be on the linear predictor scale. Specifically:

- **Binomial model** - $Y^*(\mathcal{S}_j) = \ln[\{Y^D(\mathcal{S}_j)/N(\mathcal{S}_j)\}/\{1 - Y^D(\mathcal{S}_j)/N(\mathcal{S}_j)\}]$ - the logit proportion.
- **Poisson model** - $Y^*(\mathcal{S}_j) = \ln[Y^A(\mathcal{S}_j)/N(\mathcal{S}_j)]$ - the log rate.

In both cases the asymmetric row-standardised weight matrix $\mathbf{W} = (w_{ij})$ is computed using the 6 nearest neighbours rule, where $w_{ij} = 1/6$ if LSOA $\mathcal{S}_j$ is one of the six nearest neighbours to $\mathcal{S}_i$ in terms of inter-centroidal distance, and $w_{ij} = 0$ otherwise including if $i = j$. Here, 6 nearest neighbours are chosen because it matches the average number of neighbours from the border sharing rule. Note, the border sharing rule was not used here because it results in boundary effects, where LSOAs on the edge of the study region have many fewer neighbours than those LSOAs not on the edge.

The above model was fitted separately for each mental ill health outcome, census time period and social frontier variable, using maximum quasi-likelihood estimation via the R software. Finally, we note in passing that while the social frontier metrics (1) and (2) are computed for each areal unit $\mathcal{S}_i$ and can hence be included in the regression model (3) as a covariate, this is not possible for the index of dissimilarity D because it is an overall measure of segregation for the entire study region rather than having a specific value for each area unit (see Section 1 in the supplementary material for its definition).

Table 1
Summary of the 2 social frontier metrics for each year and ethnic/religious grouping.

Quantity	Variable	Year	
		2011	2021
Mean frontier size Mean (Min, Max)	African	0.98 (0.00, 22.04)	1.35 (0.00, 34.85)
	Bangladeshi	0.57 (0.00, 39.07)	0.67 (0.00, 36.56)
	Caribbean	0.50 (0.00, 18.07)	0.50 (0.00, 15.38)
	Chinese	0.51 (0.00, 21.66)	0.51 (0.00, 21.71)
	Indian	1.31 (0.00, 46.62)	1.52 (0.00, 46.70)
	Pakistani	1.17 (0.00, 46.54)	1.42 (0.00, 43.24)
	White British	4.93 (0.03, 54.05)	5.70 (0.02, 52.77)
	Christian	5.33 (0.07, 38.23)	5.51 (0.01, 33.10)
	Hindu	0.80 (0.00, 29.48)	0.93 (0.00, 28.79)
	Jewish	0.34 (0.00, 38.34)	0.34 (0.00, 33.48)
	Muslim	2.26 (0.00, 52.83)	2.76 (0.00, 51.95)
	None	4.50 (0.12, 23.62)	5.32 (0.05, 27.83)
Maximum frontier size Mean (Min, Max)	African	2.04 (0.00, 33.69)	2.81 (0.00, 45.93)
	Bangladeshi	1.25 (0.00, 58.67)	1.48 (0.00, 57.97)
	Caribbean	1.02 (0.00, 25.82)	1.02 (0.00, 20.19)
	Chinese	1.08 (0.00, 28.86)	1.08 (0.00, 28.55)
	Indian	2.77 (0.00, 68.89)	3.20 (0.00, 68.22)
	Pakistani	2.56 (0.00, 72.27)	3.06 (0.00, 76.00)
	White British	10.35 (0.04, 94.37)	11.88 (0.03, 89.37)
	Christian	10.52 (0.11, 72.94)	10.77 (0.01, 67.27)
	Hindu	1.68 (0.00, 62.00)	1.94 (0.00, 64.88)
	Jewish	0.74 (0.00, 64.83)	0.73 (0.00, 60.68)
	Muslim	4.84 (0.00, 94.19)	5.84 (0.00, 88.35)
	None	8.79 (0.12, 38.55)	10.39 (0.05, 45.54)

4. Results of the study

The results of the study are presented below, beginning with the construction of the social frontiers in Section 4.1. The model building process used to construct the final models is presented in Section 4.2, while the effects of the social frontiers on the prevalence of depression are presented in Section 4.3.

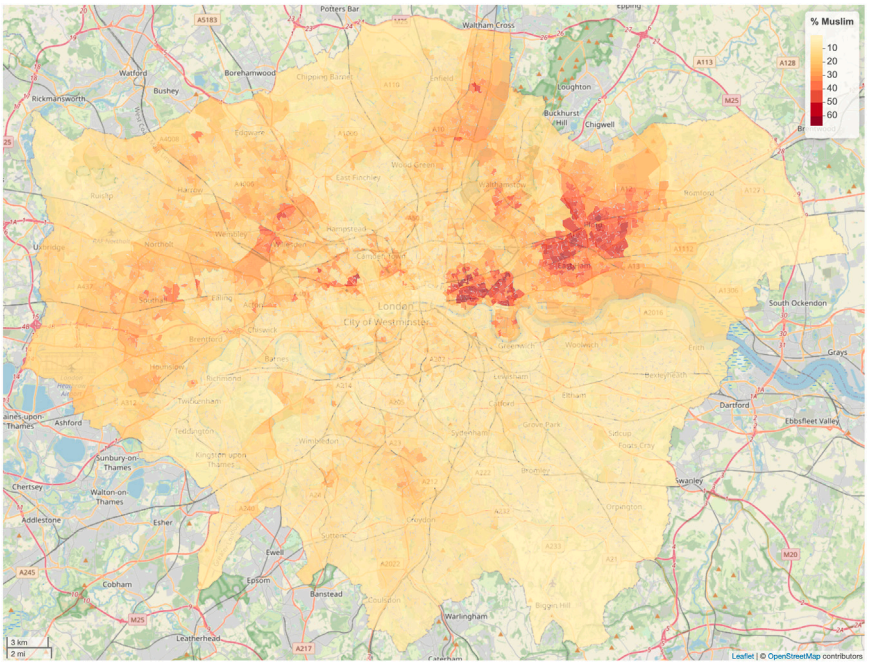
4.1. Construction of the social frontiers

Table 1 presents the mean, minimum and maximum values for all the social frontier metrics, which shows large variation in frontier sizes across England. The groups with the lowest and highest mean frontier sizes are Caribbean (0.5%) and White British (4.93% to 5.7%) for ethnicity, and Jewish (0.34%) and Christian (5.33% to 5.51%) for religion. Additionally, frontier size has either stayed roughly similar (e.g., Caribbean and Jewish) or increased (e.g., Muslim and White British) from 2011 to 2021, suggesting that social frontiers have generally become larger and more common over the 10-year period. In contrast, the levels of community segregation in respect of these categories as measured by the index of dissimilarity are largely unchanged between the two study periods, and the results of this analysis are presented in Section 3.1 of the supplementary material accompanying this paper.

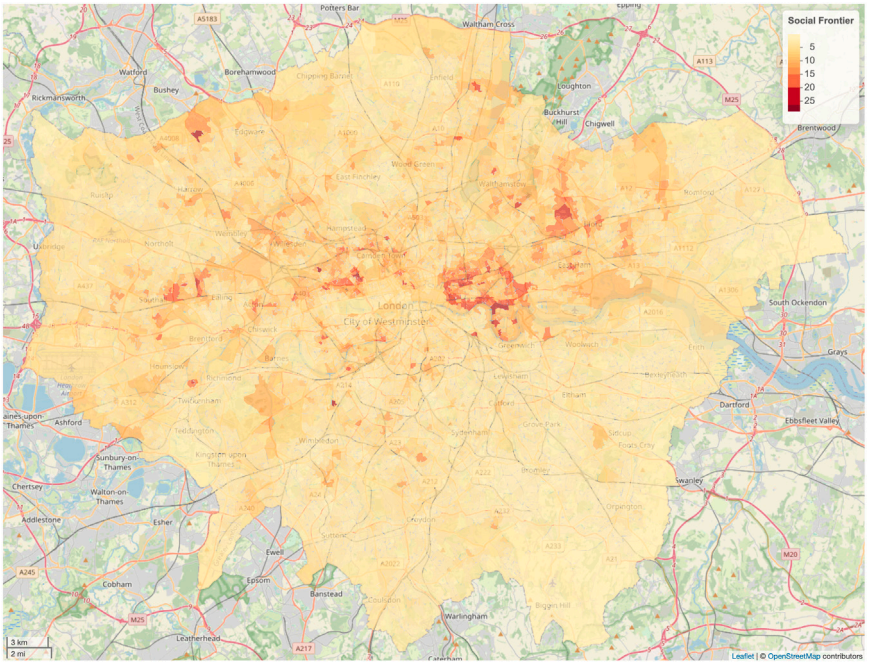
To illustrate the spatial patterning in the social frontiers metrics, Fig. 1 displays for Greater London in 2021: (a) the percentage of the population in each LSOA who are Muslim; and (b) the mean frontier size metric based on these LSOA-level Muslim percentage data. The top panel shows that the majority of the Muslim population are located in central and east London in the Tower Hamlets (central) and Newham (east) areas. The edges of these high-percentage Muslim areas are clearly demarcated in panel (b), which illustrates that the social frontier metrics are identifying these community boundaries.

4.2. Model building process

Separate binomial (depression outcome) and Poisson (antidepressant outcome) regression models are fitted for each depression outcome



(a) % of the population who are Muslim.



(b) Mean frontier size for Muslim.

Fig. 1. Maps of Greater London for 2021 displaying: (a) the percentage of the population in each LSOA who are Muslim; and (b) the mean frontier size metric based on these LSOA-level Muslim percentage data.

and time period, resulting in four sets of model results. Each model contains a single social frontier metric and an appropriate set of confounders, with the social frontier variables being included in separate models to prevent any potential confounding between multiple frontier types. This results in 96 estimated social frontier effects across all analyses, which is comprised of 2 time periods × 2 health outcomes × 2 social frontier types (mean and max frontier sizes) × 12 ethnic/religion frontier groups.

The set of confounders were assessed for collinearity by examining Pearson's correlation coefficients and variance inflation factors (VIF). For both census years the employment and income scores of the IMD had correlation coefficients of at least 0.95, resulting in the employment score being removed in all subsequent models. The remaining confounders were then included in initial models and the VIFs were computed. Confounders with VIFs above five were then removed as they are indicative of collinerity. A backwards elimination variable selection approach based on AIC was then applied to the remaining confounders, which led to the final confounder set being selected. The exceptions to this are the variables used to create the social frontier metrics (e.g., percentage Pakistani), which are retained in the

corresponding model as confounders when estimating the effect of that variable's social frontier. This inclusion ensures that the estimated social frontier effects are not confounded by the lack of the corresponding areal-level main effect for that variable in the model.

The residuals from each of these final confounder models were checked for the presence of spatial autocorrelation using Moran's I statistic (Moran, 1950), where a p-value was computed against the null hypothesis of no spatial autocorrelation using a permutation testing approach. The residuals from all models exhibited statistically significant spatial autocorrelation at the 0.1% level, with Moran's I statistics of: (a) depression diagnoses in 2011 – 0.785; (b) depression diagnoses in 2021 - 0.761; (c) antidepressant usage in 2011 – 0.723; and (d) antidepressant usage in 2021 – 0.688. As a result, all final models incorporated an additional spatially lagged response variable to account for this autocorrelation as described in Section 3.3. Although not of primary interest, the effects of the confounders on mental ill health are displayed in Section 3.2 of the supplementary material for completeness.

4.3. Associations between social frontiers and mental ill health

The associations between each social frontier type and depression outcome are estimated from the models outlined in Section 3.3, but we present additional analyses in Sections 3.3 and 3.4 of the supplementary material to illustrate the robustness of the results. In Section 3.3 this supplementary analysis compares different approaches for controlling for residual spatial autocorrelation when fitting the models, while in Section 3.4 we examine the impact of an overall frontier index aggregated over all the ethnic or religious groupings.

However, the main study results are displayed in Tables 2 and 3, which respectively relate to the ethnicity and religious social frontier metrics. The results for the depression diagnoses outcome are obtained from binomial logistic regression models, and hence the estimated effect sizes are presented as odds ratios (OR). In contrast, the results for the antidepressant usage outcome are obtained from Poisson log-linear regression models, and hence the estimated effect sizes are presented as relative risks (RR). Both ORs and RRs relate to a 5% change in the social frontier metric, i.e., what is the associated change in mental ill health if the mean or maximum social frontier size goes from 15% to 20% or from 30% to 35%. We choose this common scale of increase for ease of exposition, but appreciate that the social frontier metrics have different levels of variation across the ethnicity/religious variables as well as between the mean and max frontier sizes. For example, a 5% increase in the mean frontier size for being Jewish covers a much larger proportion of its distribution (IQR - 0.14%) than the same increase in the maximum frontier size for being Christian (IQR - 6.6%). However, we note that although the absolute size of the effect sizes will change depending on the increase chosen (e.g., 5%), the statistical significance of the effects is invariant to this choice.

The OR and RR results presented in Tables 2 and 3 comprise effect estimates and 95% confidence intervals for each social frontier type and mental ill health outcome, where statistically significant harmful effects of social frontiers at the 5% level are underlined, while significant beneficial effects are italicised. Note, as the tables contain 96 separate effect estimates we do not seek to draw conclusions on the basis of a single significant effect, as such an effect could be an artefact of the large numbers of confidence intervals constructed. Taking the results as a whole shows that social frontiers have a consistent harmful effect on the prevalence of depression, with 67% (64 out of 96) of the estimated associations being greater than one and statistically significant at the 5% level. Conversely, there is almost no evidence that social frontiers are beneficial to health, as only 3% (3 out of 96) of the OR and RR results are significantly less than one and their corresponding effect sizes are very small. Furthermore, the statistical significance (yes/no) of the individual results are largely consistent across multiple similar pairs of social frontier types and mental ill health outcomes, suggesting a clear robustness to our findings. For example, 83% of the significance levels are the same when comparing mean or maximum frontier size, 79% are the same when comparing the two census years, while 79% are the same when comparing the depression and antidepressant outcomes.

Focusing on ethnicity, consistently significant harmful effects of social frontiers on the prevalence of depression were observed for the following groupings: Indian (all 8 estimates are significant); Pakistani (8); White British (8) and Chinese (6). The average estimated effect sizes for these groups range between around a 0.2% increased odds/risk if the White British social frontier increased by 5%, compared to an average of around 2.2% increased odds/risk if the Chinese social frontier increased by 5%. The remaining ethnic groupings did not show consistent significant effects. Similarly, some religious social frontiers also show consistent significant harmful effects on the prevalence of depression, including Hindu (all 8 estimates are significant), Jewish (8) and Muslim (8). These estimated effect sizes are similar, with most ORs and RRs corresponding to a 5% increase in the frontier size being up to 1% for Muslim, up to 1.9% for Hindu and up to 1.6% for Jewish. In contrast, there is no evidence that social frontiers relating to either Christianity or having no religion have any association with the prevalence of depression.

5. Discussion

This paper presents the first comprehensive analysis of the association between the prevalence of depression and the presence of invisible social frontiers (abrupt neighbourhood borders) that are observed in many cities throughout the world. Our analysis extends previous research by Maguire et al. (2016) who found significantly higher take up of antidepressants and anxiolytic drugs near the specialised situation of Peace Walls in Northern Ireland. Our study uses a population-level spatial ecological design at the LSOA scale in England, and we assess the robustness of our findings by estimating the impact of 24 different social frontier types on two measures of depression over two time periods.

For both 2011 and 2021 we find strong and consistent evidence of a negative association between the intensity of social frontiers and the prevalence of depression. That is, for most ethnic and religious groups our results suggest that living in an area segregated by social frontiers is detrimental to mental health, with odds ratios and relative risks in the range of a 1% to 2% increase in the prevalence of depression if the size of the social frontier increases by 5%. This negative association could be due to the nature of these social frontiers that impose a strict geographical division, limiting opportunities for intergroup contact. Without intergroup contact, the involved groups miss one of the key mechanisms contributing to more positive intergroup attitudes and relations (Pettigrew, 2008). Instead, the presence of these frontiers can exacerbate negative mechanisms such as territorial effects and stronger divisions between groups, which may contribute to ingroup favouritism (Tajfel and Turner, 1978) and potential polarisation between groups.

It is important to note groups such as the Caribbean or Christians where such effects were not found. This may be due to mitigating factors. For example, research has shown that African Americans tend to more strongly identify with their racial group in the presence of discrimination and this may fully counteract the negative effect of discrimination on well-being (Branscombe et al., 1999). Also, there might be some ethnic density effects when a high in-group proportion provides protective effects on mental health (Bosqui et al., 2014; Das-Munshi et al., 2010). In our study we aimed to provide an overview of the social frontier effects across major ethnic and religious groups in England. More detailed information and analysis could be provided by future research focusing on the specific characteristics of each group and the particular relationships developed with the other group or groups at a social frontier.

Another interesting finding is that our study shows broadly similar levels of residential segregation between the two census years, as measured by the index of dissimilarity, which slightly disagrees with Catney

Table 2
Estimates and 95% confidence intervals for the effects of ethnic social frontiers on depression (OR) and antidepressant usage (RR). In each case the OR/RR relates to the change in the mental ill health outcome associated with increasing the size of the social frontier by 5 (on a percentage absolute difference scale). Underlined elements indicate harmful frontiers and italicised elements indicate beneficial frontiers that are significant at the 5% level.

Ethnic group	Year	Depression (OR)		Antidepressants (RR)	
		Mean size	Max size	Mean size	Max size
African	2011	1.009 (1.001, 1.017)	1.005 (1.002, 1.009)	0.993 (0.999, 1.004)	1.001 (0.998, 1.003)
	2021	0.999 (0.995, 1.003)	1.000 (0.999, 1.002)	0.998 (0.995, 1.002)	0.999 (0.998, 1.001)
Bangladeshi	2011	1.002 (0.995, 1.010)	1.003 (1.000, 1.006)	1.006 (1.0012, 1.011)	1.004 (1.002, 1.005)
	2021	0.999 (0.995, 1.004)	1.002 (1.000, 1.003)	1.002 (0.998, 1.005)	1.002 (1.000, 1.003)
Caribbean	2011	1.014 (0.999, 1.030)	1.008 (1.001, 1.015)	1.007 (0.996, 1.017)	1.002 (0.997, 1.007)
	2021	0.998 (0.988, 1.008)	0.999 (0.994, 1.003)	0.993 (0.984, 1.003)	0.999 (0.995, 1.003)
Chinese	2011	1.015 (0.998, 1.032)	1.017 (1.012, 1.023)	1.007 (0.995, 1.018)	1.027 (1.024, 1.031)
	2021	1.022 (1.011, 1.033)	1.015 (1.011, 1.018)	1.044 (1.034, 1.054)	1.023 (1.020, 1.027)
Indian	2011	1.008 (1.003, 1.013)	1.006 (1.004, 1.008)	1.011 (1.007, 1.014)	1.007 (1.005, 1.008)
	2021	1.010 (1.007, 1.013)	1.006 (1.005, 1.007)	1.011 (1.008, 1.014)	1.007 (1.005, 1.008)
Pakistani	2011	1.016 (1.012, 1.021)	1.009 (1.007, 1.010)	1.012 (1.010, 1.015)	1.006 (1.005, 1.007)
	2021	1.009 (1.006, 1.011)	1.005 (1.004, 1.006)	1.008 (1.006, 1.010)	1.004 (1.003, 1.005)
White British	2011	1.002 (1.001, 1.004)	1.003 (1.002, 1.003)	1.002 (1.001, 1.003)	1.002 (1.002, 1.003)
	2021	1.001 (1.000, 1.002)	1.001 (1.001, 1.002)	1.003 (1.002, 1.004)	1.002 (1.002, 1.002)

Table 3
Estimates and 95% confidence intervals for the effects of the religion social frontiers on depression diagnoses presented as odds ratios (OR) and antidepressant usage presented as relative risks (RR). In each case the OR/RR relates to the change in the mental ill health outcome associated with increasing the size of the social frontier by 5 (on a percentage absolute difference scale). Underlined elements indicate harmful frontiers and italicised elements indicate beneficial frontiers that are significant at the 5% level.

Religion	Year	Depression (OR)		Antidepressants (RR)	
		Mean size	Max size	Mean size	Max size
Christian	2011	1.001 (0.998, 1.003)	1.002 (1.001, 1.004)	*0.998 (0.996, 0.999)*	1.001 (1.000, 1.001)
	2021	1.000 (0.999, 1.002)	1.001 (1.000, 1.001)	0.999 (0.998, 1.000)	1.000 (0.999, 1.001)
Hindu	2011	1.019 (1.010, 1.027)	1.011 (1.008, 1.015)	1.016 (1.010, 1.021)	1.010 (1.008, 1.012)
	2021	1.013 (1.008, 1.019)	1.007 (1.005, 1.010)	1.011 (1.006, 1.016)	1.008 (1.006, 1.009)
Jewish	2011	1.013 (1.003, 1.022)	1.007 (1.003, 1.010)	1.014 (1.009, 1.020)	1.006 (1.004, 1.008)
	2021	1.016 (1.010, 1.022)	1.007 (1.005, 1.010)	1.015 (1.010, 1.021)	1.006 (1.004, 1.008)
Muslim	2011	1.010 (1.007, 1.012)	1.006 (1.004, 1.007)	1.009 (1.008, 1.011)	1.004 (1.004, 1.005)
	2021	1.005 (1.003, 1.006)	1.003 (1.002, 1.004)	1.006 (1.005, 1.008)	1.003 (1.003, 1.004)
None	2011	0.999 (0.996, 1.002)	1.000 (0.999, 1.002)	*0.997 (0.995, 0.999)*	*0.999 (0.998, 1.000)*
	2021	1.000 (0.998, 1.002)	1.000 (1.000, 1.001)	1.002 (1.000, 1.003)	1.001 (1.000, 1.002)

(2018) who find declining levels of segregation over time. In addition, we find that the sizes of most social frontiers have increased from 2011 to 2021. Thus, if social frontiers have a significant association with the prevalence of depression as our results suggest, this is a significant omission from the segregation literature and raises the imperative to develop appropriate aggregate measures that capture the overall level of social frontier prevalence in a city or region.

We have focused on establishing whether there is an association between ethnic and religious social frontiers and the prevalence of depression, and the results in our study suggest several interesting avenues for future research. The first is to investigate whether frontiers in other attributes, such as social class, country of birth or political affiliation, may also have the potential to evoke territorial behaviour or be a source of inter-group tension, and hence be deleterious for the mental health of nearby communities. On a related theme, intersectional social frontiers, that is abrupt transitions in multiple combinations of group identity (Iyer and Pryce, 2024), may also be an important area for future investigation. Secondly, a natural extension of the present study is to develop causal inference strategies to identify the direction and magnitude of the causal relationship. Thirdly, more research is also needed on the factors that cause social frontiers to emerge, such as social housing allocation decisions and town planning. The latter may be especially salient because of the tendency for neighbourhood boundaries to follow existing physical barriers such as main roads, waste-land, railways, parks and rivers (Noonan, 2005). Moreover, infrastructure and planning decisions may be manipulated by the majority group to deliberately impose *'barricades'* (Sharkey, 2020) that help maintain a degree of physical and perceived distance from the minority group, and minimise the flows of people between the two communities. Our results provide the first set of evidence across multiple urban areas that such separation of groups might have implications for mental health, which suggests that planners need to be cognisant of the social cohesion impacts of their decisions.

CRediT authorship contribution statement

Duncan Lee: Writing – review & editing, Methodology, Formal analysis, Data curation. **Gwilym Pryce:** Writing – original draft, Investigation, Conceptualization. **Miguel R. Ramos:** Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Prof Pryce reports financial support was provided by the UK Research and Innovation Economic and Social Research Council. Prof Lee is on the editorial board of Spatial and Spatio-temporal Epidemiology. There are no other conflicts of interest or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.sste.2026.100854.

Data availability

All data are publicly available and the sources are given in the manuscript.

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